

**BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI, DUBAI CAMPUS**  
**DUBAI INTERNATIONAL ACADEMIC CITY, DUBAI**  
**SECOND SEMESTER 2023 – 2024**  
**DEPARTMENT OF COMPUTER SCIENCE**

**Questions: 5**  
**Pages: 3**

**Question Paper**

**COURSE :** Machine Learning (BITS F464)

**COMPONENT :** Midsem Exam CLOSED BOOK      **WEIGHTAGE :** 30% (30 Marks)

**DATE :** 1-April-2024      **DURATION :** 90 Minutes (8:30 am- 10:00 am)

**Note:**

1. Please write down any assumptions that you make.
2. Show the intermittant steps clearly for all answers.

| Q. 1                       | <div>Consider a following dataset which shows willingness of buying insurance by considering Gender, income, and credit score,</div> <div>Table 1: Dataset of Willingness of Buying Insurance</div> <table><tr><th>Gender</th><th>Income</th><th>Credit Score</th><th>Willingness of Buying Insurance</th></tr><tr><td>Male</td><td>High</td><td>Excellent</td><td>Yes</td></tr><tr><td>Female</td><td>High</td><td>Fair</td><td>No</td></tr><tr><td>Male</td><td>High</td><td>Excellent</td><td>Yes</td></tr><tr><td>Female</td><td>Medium</td><td>Excellent</td><td>Yes</td></tr><tr><td>Male</td><td>Low</td><td>Fair</td><td>No</td></tr><tr><td>Female</td><td>Low</td><td>Poor</td><td>No</td></tr><tr><td>Male</td><td>High</td><td>Poor</td><td>Yes</td></tr></table> <div><div>i.</div><div>Construct the decision tree using Gini Index and show all steps of calculations.</div></div> <div><div>ii.</div><div>Find the threshold value for converting following credit score continuous values to discrete for constructing decision tree using Gini Index for classifying samples into Yes/No. (Show all steps of calculation.)</div></div> <table><tr><td>Credit Score</td><td>53</td><td>59</td><td>71</td><td>83</td><td>91</td><td>101</td></tr></table> | Gender                     | Income                          | Credit Score | Willingness of Buying Insurance | Male | High | Excellent | Yes | Female | High | Fair                  | No  | Male | High | Excellent | Yes | Female | Medium | Excellent | Yes | Male | Low | Fair | No | Female | Low | Poor | No | Male | High | Poor | Yes | Credit Score | 53 | 59 | 71 | 83 | 91 | 101 | 5 + 2 M |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|---------------------------------|--------------|---------------------------------|------|------|-----------|-----|--------|------|-----------------------|-----|------|------|-----------|-----|--------|--------|-----------|-----|------|-----|------|----|--------|-----|------|----|------|------|------|-----|--------------|----|----|----|----|----|-----|---------|
| Gender                     | Income                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Credit Score               | Willingness of Buying Insurance |              |                                 |      |      |           |     |        |      |                       |     |      |      |           |     |        |        |           |     |      |     |      |    |        |     |      |    |      |      |      |     |              |    |    |    |    |    |     |         |
| Male                       | High                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Excellent                  | Yes                             |              |                                 |      |      |           |     |        |      |                       |     |      |      |           |     |        |        |           |     |      |     |      |    |        |     |      |    |      |      |      |     |              |    |    |    |    |    |     |         |
| Female                     | High                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Fair                       | No                              |              |                                 |      |      |           |     |        |      |                       |     |      |      |           |     |        |        |           |     |      |     |      |    |        |     |      |    |      |      |      |     |              |    |    |    |    |    |     |         |
| Male                       | High                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Excellent                  | Yes                             |              |                                 |      |      |           |     |        |      |                       |     |      |      |           |     |        |        |           |     |      |     |      |    |        |     |      |    |      |      |      |     |              |    |    |    |    |    |     |         |
| Female                     | Medium                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Excellent                  | Yes                             |              |                                 |      |      |           |     |        |      |                       |     |      |      |           |     |        |        |           |     |      |     |      |    |        |     |      |    |      |      |      |     |              |    |    |    |    |    |     |         |
| Male                       | Low                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Fair                       | No                              |              |                                 |      |      |           |     |        |      |                       |     |      |      |           |     |        |        |           |     |      |     |      |    |        |     |      |    |      |      |      |     |              |    |    |    |    |    |     |         |
| Female                     | Low                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Poor                       | No                              |              |                                 |      |      |           |     |        |      |                       |     |      |      |           |     |        |        |           |     |      |     |      |    |        |     |      |    |      |      |      |     |              |    |    |    |    |    |     |         |
| Male                       | High                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Poor                       | Yes                             |              |                                 |      |      |           |     |        |      |                       |     |      |      |           |     |        |        |           |     |      |     |      |    |        |     |      |    |      |      |      |     |              |    |    |    |    |    |     |         |
| Credit Score               | 53                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 59                         | 71                              | 83           | 91                              | 101  |      |           |     |        |      |                       |     |      |      |           |     |        |        |           |     |      |     |      |    |        |     |      |    |      |      |      |     |              |    |    |    |    |    |     |         |
| Q. 2                       | <div>A company manufactures an electronic device to be used in a very wide temperature range. The company knows that increased temperature shortens the life time of the device, and a study is therefore performed in which the life time is determined as a function of temperature. The following data is found,</div> <div>Table 2: Dataset for temperature and lifetime in hours for electronic devices</div> <table><tr><td>Temperature in Celcius (t)</td><td>10</td><td>20</td><td>30</td><td>40</td><td>50</td><td>60</td><td>70</td><td>80</td><td>90</td></tr><tr><td>Lifetime in hours (y)</td><td>420</td><td>365</td><td>285</td><td>220</td><td>176</td><td>117</td><td>69</td><td>34</td><td>5</td></tr></table> <div><div>i.</div><div>Estimate the equation of regression line <math>y = mx + c</math>, for given data by considering temperature in celcius as independent variable and lifetime in hour as dependent variable. Show all steps of calculation. (3 M)</div></div> <div><div>ii.</div><div>Draw the regression line that best fit the given data. Show all steps of calculation. (1 M)</div></div> <div><div>iii.</div><div>Calculate the <math>R^2</math> value for given data. Show all steps of calculation. (1 M)</div></div>        | Temperature in Celcius (t) | 10                              | 20           | 30                              | 40   | 50   | 60        | 70  | 80     | 90   | Lifetime in hours (y) | 420 | 365  | 285  | 220       | 176 | 117    | 69     | 34        | 5   | 5 M  |     |      |    |        |     |      |    |      |      |      |     |              |    |    |    |    |    |     |         |
| Temperature in Celcius (t) | 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 20                         | 30                              | 40           | 50                              | 60   | 70   | 80        | 90  |        |      |                       |     |      |      |           |     |        |        |           |     |      |     |      |    |        |     |      |    |      |      |      |     |              |    |    |    |    |    |     |         |
| Lifetime in hours (y)      | 420                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 365                        | 285                             | 220          | 176                             | 117  | 69   | 34        | 5   |        |      |                       |     |      |      |           |     |        |        |           |     |      |     |      |    |        |     |      |    |      |      |      |     |              |    |    |    |    |    |     |         |

| Q. 3             | <p>Consider the following dataset of two species of Iris flower shown in Table 3. Apply the Linear Discriminant Analysis (LDA) and plot the samples in reduced dimension. Show all steps of calculation.</p> <p style="text-align: center;"><b>Table 3: Iris Flower Dataset</b></p> <table><tr><th>Sepal Length</th><th>Sepal Width</th><th>Class</th></tr><tr><td>5.3</td><td>3.7</td><td>Setosa</td></tr><tr><td>5.1</td><td>3.8</td><td>Setosa</td></tr><tr><td>7.2</td><td>3.0</td><td>Virginica</td></tr><tr><td>5.4</td><td>3.4</td><td>Setosa</td></tr><tr><td>7.4</td><td>2.8</td><td>Virginica</td></tr><tr><td>7.3</td><td>2.9</td><td>Virginica</td></tr><tr><td>5.0</td><td>3.2</td><td>Setosa</td></tr></table>                                                                                                                                                                                                                                        | Sepal Length     | Sepal Width | Class         | 5.3 | 3.7 | Setosa | 5.1 | 3.8 | Setosa | 7.2 | 3.0 | Virginica | 5.4 | 3.4 | Setosa | 7.4 | 2.8 | Virginica | 7.3 | 2.9 | Virginica | 5.0 | 3.2 | Setosa | 8 M |   |    |   |   |    |     |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-------------|---------------|-----|-----|--------|-----|-----|--------|-----|-----|-----------|-----|-----|--------|-----|-----|-----------|-----|-----|-----------|-----|-----|--------|-----|---|----|---|---|----|-----|
| Sepal Length     | Sepal Width                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Class            |             |               |     |     |        |     |     |        |     |     |           |     |     |        |     |     |           |     |     |           |     |     |        |     |   |    |   |   |    |     |
| 5.3              | 3.7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Setosa           |             |               |     |     |        |     |     |        |     |     |           |     |     |        |     |     |           |     |     |           |     |     |        |     |   |    |   |   |    |     |
| 5.1              | 3.8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Setosa           |             |               |     |     |        |     |     |        |     |     |           |     |     |        |     |     |           |     |     |           |     |     |        |     |   |    |   |   |    |     |
| 7.2              | 3.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Virginica        |             |               |     |     |        |     |     |        |     |     |           |     |     |        |     |     |           |     |     |           |     |     |        |     |   |    |   |   |    |     |
| 5.4              | 3.4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Setosa           |             |               |     |     |        |     |     |        |     |     |           |     |     |        |     |     |           |     |     |           |     |     |        |     |   |    |   |   |    |     |
| 7.4              | 2.8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Virginica        |             |               |     |     |        |     |     |        |     |     |           |     |     |        |     |     |           |     |     |           |     |     |        |     |   |    |   |   |    |     |
| 7.3              | 2.9                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Virginica        |             |               |     |     |        |     |     |        |     |     |           |     |     |        |     |     |           |     |     |           |     |     |        |     |   |    |   |   |    |     |
| 5.0              | 3.2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Setosa           |             |               |     |     |        |     |     |        |     |     |           |     |     |        |     |     |           |     |     |           |     |     |        |     |   |    |   |   |    |     |
| Q. 4             | <p>Consider a following confusion matrices as shown in Table 4 for classification of samples into 4 different classes.</p> <p style="text-align: center;"><b>Table 4: Confusion Matrix</b></p> <table><tr><th rowspan="5">Predicted values</th><th rowspan="2"></th><th colspan="4">Actual values</th></tr><tr><th>A</th><th>B</th><th>C</th><th>D</th></tr><tr><th>A</th><td>10</td><td>2</td><td>1</td><td>2</td></tr><tr><th>B</th><td>5</td><td>12</td><td>1</td><td>2</td></tr><tr><th>C</th><td>5</td><td>2</td><td>18</td><td>10</td></tr><tr><th>D</th><td>11</td><td>3</td><td>1</td><td>15</td></tr></table> <p>Calculate the total and respective (individual) F1-score and accuracy for class A, B, C, and D. Show all steps of calculation.</p>                                                                                                                                                                                                        | Predicted values |             | Actual values |     |     |        | A   | B   | C      | D   | A   | 10        | 2   | 1   | 2      | B   | 5   | 12        | 1   | 2   | C         | 5   | 2   | 18     | 10  | D | 11 | 3 | 1 | 15 | 5 M |
| Predicted values |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                  |             | Actual values |     |     |        |     |     |        |     |     |           |     |     |        |     |     |           |     |     |           |     |     |        |     |   |    |   |   |    |     |
|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                  | A           | B             | C   | D   |        |     |     |        |     |     |           |     |     |        |     |     |           |     |     |           |     |     |        |     |   |    |   |   |    |     |
|                  | A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                  | 10          | 2             | 1   | 2   |        |     |     |        |     |     |           |     |     |        |     |     |           |     |     |           |     |     |        |     |   |    |   |   |    |     |
|                  | B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                  | 5           | 12            | 1   | 2   |        |     |     |        |     |     |           |     |     |        |     |     |           |     |     |           |     |     |        |     |   |    |   |   |    |     |
|                  | C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 5                | 2           | 18            | 10  |     |        |     |     |        |     |     |           |     |     |        |     |     |           |     |     |           |     |     |        |     |   |    |   |   |    |     |
| D                | 11                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 3                | 1           | 15            |     |     |        |     |     |        |     |     |           |     |     |        |     |     |           |     |     |           |     |     |        |     |   |    |   |   |    |     |
| Q. 5             | <p>i. How many two class classifiers are required to classify samples into 5 different classes using one-vs-one approach?</p> <p>a. 12</p> <p>b. 10</p> <p>c. 5</p> <p>d. 2</p> <p>ii. There are two boxes. The first box contains 3 white and 2 red balls whereas the second contains 5 white and 4 red balls. A ball is drawn at random from one of the two boxes and is found to be white. Find the probability that the ball was drawn from the second box?</p> <p>a. 53/50</p> <p>b. 50/104</p> <p>c. 54/104</p> <p>d. 54/44</p> <p>iii. Which of the following options is/are true for K-fold cross-validation?</p> <p>1. Increase in K will result in higher time required to cross validate the result.</p> <p>2. Higher values of K will result in higher confidence on the cross-validation result as compared to lower value of K.</p> <p>3. Decrease in K will result in higher time required to cross validate the result.</p> <p>a. All of these.</p> | 1 M*5            |             |               |     |     |        |     |     |        |     |     |           |     |     |        |     |     |           |     |     |           |     |     |        |     |   |    |   |   |    |     |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | <ul style="list-style-type: none"> <li>b. 1 and 2</li> <li>c. 2 and 3</li> <li>d. 1 and 3</li> </ul> <p>iv. Imagine you are using a k-Nearest Neighbor classifier on a dataset with lots of noise. You want your classifier to be less sensitive to the noise. Which of the following is likely to help and with what side effect?</p> <ul style="list-style-type: none"> <li>a. Increase the value of k → Increase in prediction time</li> <li>b. Decrease the value of k → Increase in prediction time</li> <li>c. Increase the value of k → Decrease in prediction time</li> <li>d. Decrease the value of k → Decrease in prediction time</li> </ul> <p>v. Consider a scenario where you are applying LASSO Regression to a dataset with highly correlated features. How does the introduction of L1 regularization affect the behavior of the model in terms of selecting features?</p> <ul style="list-style-type: none"> <li>a. It tends to select only one of the correlated features and ignore the others</li> <li>b. It tends to select all correlated features</li> <li>c. It tends to drop all correlated features</li> <li>d. It tends to increase the correlation between features</li> </ul> |  |
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\*\*\*\*\* ALL THE BEST \*\*\*\*\*

**BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI, DUBAI CAMPUS**  
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**SECOND SEMESTER 2023 – 2024**  
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**Marking Scheme**

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**DATE :** 1-April-2024

**DURATION :** 90 Minutes (8:30 am- 10:00 am)

**Note:**

1. Please write down any assumptions that you make.
2. Show the intermittant steps clearly for all answers.

**Q. 1**

Consider a following dataset which shows willingness of buying insurance by considering Gender, income, and credit score,

**Table 1: Dataset of Willingness of Buying Insurance**

| Gender | Income | Credit Score | Willingness of Buying Insurance |
|--------|--------|--------------|---------------------------------|
| Male   | High   | Excellent    | Yes                             |
| Female | High   | Fair         | No                              |
| Male   | High   | Excellent    | Yes                             |
| Female | Medium | Excellent    | Yes                             |
| Male   | Low    | Fair         | No                              |
| Female | Low    | Poor         | No                              |
| Male   | High   | Poor         | Yes                             |

- i. Construct the decision tree using Gini Index and show all steps of calculations.
- ii. Find the threshold value for converting following credit score continuous values to discrete for constructing decision tree using Gini Index for classifying samples into Yes/No. (Show all steps of calculation.)

|                     |    |    |    |    |    |     |
|---------------------|----|----|----|----|----|-----|
| <b>Credit Score</b> | 53 | 59 | 71 | 83 | 91 | 101 |
|---------------------|----|----|----|----|----|-----|

**5 + 2 M**

**Ans**

Calculation of correct root node and splitting of data. Credit score is root node => 2 M

Further induction of subtrees => 2 M

Correct decision tree => 1 M

Correct threshold value with calculations => 2 M

**Q. 2**

A company manufactures an electronic device to be used in a very wide temperature range. The company knows that increased temperature shortens the life time of the device, and a study is therefore performed in which the life time is determined as a function of temperature. The following data is found,

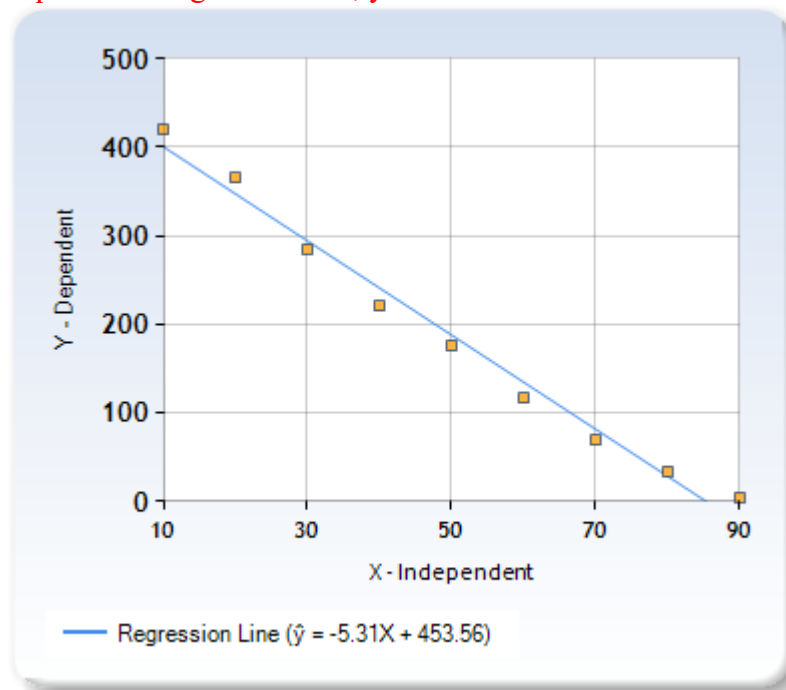
**5 M****Table 2: Dataset for temperature and lifetime in hours for electronic devices**

| Temperature in Celcius (t) | 10  | 20  | 30  | 40  | 50  | 60  | 70 | 80 | 90 |
|----------------------------|-----|-----|-----|-----|-----|-----|----|----|----|
| Lifetime in hours (y)      | 420 | 365 | 285 | 220 | 176 | 117 | 69 | 34 | 5  |

- Estimate the equation of regression line  $y = mx + c$ , for given data by considering temperature in celcius as independent variable and lifetime in hour as dependent variable. Show all steps of calculation. (3 M)
- Draw the regression line that best fit the given data. Show all steps of calculation. (1 M)
- Calculate the  $R^2$  value for given data. Show all steps of calculation. (1 M)

**Ans**

Equation of regression line,  $\hat{y} = -5.31333X + 453.55556$



$R^2 = 1 - RSE = -16.47$

**Q. 3**

Consider the following dataset of two species of Iris flower shown in Table 3. Apply the Linear Discriminant Analysis (LDA) and plot the samples in reduced dimension. Show all steps of calculation.

**8 M****Table 3: Iris Flower Dataset**

| Sepal Length | Sepal Width | Class     |
|--------------|-------------|-----------|
| 5.3          | 3.7         | Setosa    |
| 5.1          | 3.8         | Setosa    |
| 7.2          | 3.0         | Virginica |
| 5.4          | 3.4         | Setosa    |
| 7.4          | 2.8         | Virginica |
| 7.3          | 2.9         | Virginica |
| 5.0          | 3.2         | Setosa    |

| <b>Ans</b>       | $\mu_1 = \frac{5.2}{3.525}, \mu_2 = \frac{7.3}{2.9} \Rightarrow \mathbf{1M}$<br>$S_w = \begin{pmatrix} 0.043 & 0 \\ 0 & 0.08 \end{pmatrix} \Rightarrow \mathbf{1M}$<br>$S_B = \begin{pmatrix} 4.41 & -1.26 \\ -1.26 & 0.36 \end{pmatrix} \Rightarrow \mathbf{1M}$<br>$\lambda_1 = 0, \lambda_2 = 114.75 \Rightarrow \mathbf{1M}$<br>Calculating eigen vector and normalized eigen vector $\Rightarrow \mathbf{2M}$<br>LDA classifier, $LDA\ Score = A_1v_1 + A_2v_2 \Rightarrow \mathbf{1M}$<br>Plotting of sample points in one dimension $\Rightarrow \mathbf{1M}$                                                                                                                                                                                                                         |                  |    |               |    |  |  |   |   |   |   |  |   |    |   |   |   |  |   |   |    |   |   |  |   |   |   |    |    |  |   |    |   |   |    |            |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|----|---------------|----|--|--|---|---|---|---|--|---|----|---|---|---|--|---|---|----|---|---|--|---|---|---|----|----|--|---|----|---|---|----|------------|
| <b>Q. 4</b>      | Consider a following confusion matrices as shown in Table 4 for classification of samples into 4 different classes.<br><div style="text-align: center;"><b>Table 4: Confusion Matrix</b></div> <table><tr><th rowspan="2">Predicted values</th><th rowspan="2"></th><th colspan="4">Actual values</th></tr><tr><th>A</th><th>B</th><th>C</th><th>D</th></tr><tr><td></td><th>A</th><td>10</td><td>2</td><td>1</td><td>2</td></tr><tr><td></td><th>B</th><td>5</td><td>12</td><td>1</td><td>2</td></tr><tr><td></td><th>C</th><td>5</td><td>2</td><td>18</td><td>10</td></tr><tr><td></td><th>D</th><td>11</td><td>3</td><td>1</td><td>15</td></tr></table><br>Calculate the total and respective (individual) F1-score and accuracy for class A, B, C, and D. Show all steps of calculation. | Predicted values |    | Actual values |    |  |  | A | B | C | D |  | A | 10 | 2 | 1 | 2 |  | B | 5 | 12 | 1 | 2 |  | C | 5 | 2 | 18 | 10 |  | D | 11 | 3 | 1 | 15 | <b>5 M</b> |
| Predicted values |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                  |    | Actual values |    |  |  |   |   |   |   |  |   |    |   |   |   |  |   |   |    |   |   |  |   |   |   |    |    |  |   |    |   |   |    |            |
|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | A                | B  | C             | D  |  |  |   |   |   |   |  |   |    |   |   |   |  |   |   |    |   |   |  |   |   |   |    |    |  |   |    |   |   |    |            |
|                  | A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 10               | 2  | 1             | 2  |  |  |   |   |   |   |  |   |    |   |   |   |  |   |   |    |   |   |  |   |   |   |    |    |  |   |    |   |   |    |            |
|                  | B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 5                | 12 | 1             | 2  |  |  |   |   |   |   |  |   |    |   |   |   |  |   |   |    |   |   |  |   |   |   |    |    |  |   |    |   |   |    |            |
|                  | C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 5                | 2  | 18            | 10 |  |  |   |   |   |   |  |   |    |   |   |   |  |   |   |    |   |   |  |   |   |   |    |    |  |   |    |   |   |    |            |
|                  | D                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 11               | 3  | 1             | 15 |  |  |   |   |   |   |  |   |    |   |   |   |  |   |   |    |   |   |  |   |   |   |    |    |  |   |    |   |   |    |            |
| <b>Ans</b>       | 1 mark each for correct calculation steps and F1-score and accuracy<br><ul style="list-style-type: none"><li>• Total F1-Score = 0.55, Total Accuracy = 0.55</li><li>• For Class A, F1-Score = 0.435, Accuracy = 0.74</li><li>• For Class B, F1-Score = 0.616, Accuracy = 0.85</li><li>• For Class C, F1-Score = 0.643, Accuracy = 0.81</li><li>• For Class D, F1-Score = 0.508, Accuracy = 0.71</li></ul> <div><math display="block">Accuracy = \frac{TP + TN}{(TP + TN + FP + FN)}</math><math display="block">F1\text{-score} = \frac{2}{\frac{1}{recall} + \frac{1}{precision}} \quad \text{or} \quad 2 \times (p_r \times p_c) / (p_r + p_c)</math></div>                                                                                                                                |                  |    |               |    |  |  |   |   |   |   |  |   |    |   |   |   |  |   |   |    |   |   |  |   |   |   |    |    |  |   |    |   |   |    |            |
| <b>Q. 5</b>      | <p>i. How many two class classifiers are required to classify samples into 5 different classes using one-vs-one approach?</p> <p>a. 12<br/><b>b. 10</b><br/>c. 5<br/>d. 2</p> <p>ii. There are two boxes. The first box contains 3 white and 2 red balls whereas the second contains 5 white and 4 red balls. A ball is drawn at random from one of the two boxes and is found to be white. Find the probability that the ball was drawn from the second box?</p> <p>a. 53/50<br/><b>b. 50/104</b><br/>c. 54/104</p>                                                                                                                                                                                                                                                                         | <b>1 M*5</b>     |    |               |    |  |  |   |   |   |   |  |   |    |   |   |   |  |   |   |    |   |   |  |   |   |   |    |    |  |   |    |   |   |    |            |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | <p>d. 54/44</p> <p>iii. Which of the following options is/are true for K-fold cross-validation?</p> <ol style="list-style-type: none"> <li>1. Increase in K will result in higher time required to cross validate the result.</li> <li>2. Higher values of K will result in higher confidence on the cross-validation result as compared to lower value of K.</li> <li>3. Decrease in K will result in higher time required to cross validate the result.</li> <li>a. All of these.</li> <li><b>b. 1 and 2</b></li> <li>c. 2 and 3</li> <li>d. 1 and 3</li> </ol> <p>iv. Imagine you are using a k-Nearest Neighbor classifier on a dataset with lots of noise. You want your classifier to be less sensitive to the noise. Which of the following is likely to help and with what side effect?</p> <ol style="list-style-type: none"> <li><b>a. Increase the value of k → Increase in prediction time</b></li> <li>b. Decrease the value of k → Increase in prediction time</li> <li>c. Increase the value of k → Decrease in prediction time</li> <li>d. Decrease the value of k → Decrease in prediction time</li> </ol> <p>v. Consider a scenario where you are applying LASSO Regression to a dataset with highly correlated features. How does the introduction of L1 regularization affect the behavior of the model in terms of selecting features?</p> <ol style="list-style-type: none"> <li>a. It tends to select only one of the correlated features and ignore the others</li> <li>b. It tends to select all correlated features</li> <li><b>c. It tends to drop all correlated features</b></li> <li>d. It tends to increase the correlation between features</li> </ol> |  |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

\*\*\*\*\* ALL THE BEST \*\*\*\*\*

**BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI, DUBAI CAMPUS**  
**DUBAI INTERNATIONAL ACADEMIC CITY, DUBAI**  
**SECOND SEMESTER 2023 – 2024**  
**DEPARTMENT OF COMPUTER SCIENCE**

**Questions: 7**  
**Pages: 3**

**Question Paper**

**COURSE :** Machine Learning (BITS F464)

**COMPONENT :** Endsem Exam CLOSED BOOK

**WEIGHTAGE :** 40% (40 Marks)

**DATE :** 30-May-2024

**DURATION :** 3 hrs (8:30 am- 11:30 am)

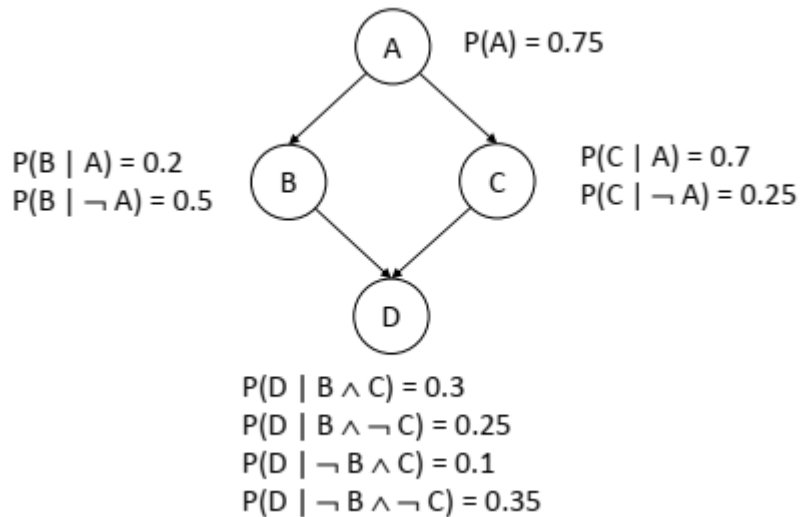
**Note:**

1. Please write down any assumptions that you make.
2. Show the intermittant steps clearly for all answers.

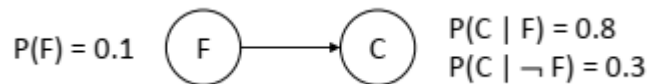
| <b>Q.1</b> | Find the cluster after two iterations of K-means clustering for the following eight points (with $(x, y)$ representing locations) into two clusters: $A1(185, 72)$ , $A2(170, 56)$ , $A3(168, 60)$ , $A4(179, 68)$ , $A5(182, 72)$ , $A6(188, 77)$ . Initial cluster centers are: $C1(185, 72)$ , and $C2(170, 56)$ . Use distance measure as Chebyshev Distance. Also find the initial mean, variance, and mixing coefficients for initial Gaussians to be used for EM for Gaussian mixture clustering. (Show all steps of calculations)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>9 M</b>                   |     |     |     |                              |                              |     |                              |                              |  |     |     |     |            |            |          |                   |                   |            |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|-----|-----|-----|------------------------------|------------------------------|-----|------------------------------|------------------------------|--|-----|-----|-----|------------|------------|----------|-------------------|-------------------|------------|
| <b>Q.2</b> | <p>Suppose we want to model Exam as decision problem, which is deciding if you will pass an exam with good marks. Student is attempting to classify the exam as either Easy or Hard. We decided to model this problem using Hidden Markov Model. Let <math>c</math> be the state in which Student classifies the Exam as Hard, and <math>h</math> be the state in which Student classifies the Exam as Easy. Rather than using legitimate techniques, Student classifies the problem based on how he classified it the day before. Between each day, there is a 70% chance that Student changes his classification of the Exam, otherwise his classification stays the same as the previous day. (Show all steps of calculations)</p> <p>a. Fill in the table with the transition probabilities as defined above, where the row corresponds to <math>X_{t+1}</math> and the column to <math>X_t</math>.</p> <table border="1" style="margin: 10px auto; border-collapse: collapse;"> <thead> <tr> <th></th><th><math>c</math></th><th><math>h</math></th></tr> </thead> <tbody> <tr> <th><math>c</math></th><td><math>P(X_{t+1} = c   X_t = c) =</math></td><td><math>P(X_{t+1} = c   X_t = h) =</math></td></tr> <tr> <th><math>h</math></th><td><math>P(X_{t+1} = h   X_t = c) =</math></td><td><math>P(X_{t+1} = h   X_t = h) =</math></td></tr> </tbody> </table> <p>b. Now suppose there is evidence in the form of Student watching outside the window. If Student classifies the Exam as Easy then there is a 60% chance he is watching outside the window. If Student classifies the Exam as Hard then there is an 80% chance he is watching outside the window. Denote watching outside the window as <math>m</math> and not watching outside window as <math>\neg m</math>. Fill in the table with the conditional probabilities of the evidence based on the description above.</p> <table border="1" style="margin: 10px auto; border-collapse: collapse;"> <thead> <tr> <th></th><th><math>c</math></th><th><math>h</math></th></tr> </thead> <tbody> <tr> <th><math>m</math></th><td><math>P(m c) =</math></td><td><math>P(m h) =</math></td></tr> <tr> <th><math>\neg m</math></th><td><math>P(\neg m   c) =</math></td><td><math>P(\neg m   h) =</math></td></tr> </tbody> </table> <p>c. What is the probability of producing the exact sequence <math>C, H, H, C, H</math> (from <math>t = 0</math> to <math>t = 4</math>)? Ignore evidence. Consider that there's a 50% chance Student's initial classification is Easy, <math>P(X_0 = c) = 0.5</math>.</p> |                              | $c$ | $h$ | $c$ | $P(X_{t+1} = c   X_t = c) =$ | $P(X_{t+1} = c   X_t = h) =$ | $h$ | $P(X_{t+1} = h   X_t = c) =$ | $P(X_{t+1} = h   X_t = h) =$ |  | $c$ | $h$ | $m$ | $P(m c) =$ | $P(m h) =$ | $\neg m$ | $P(\neg m   c) =$ | $P(\neg m   h) =$ | <b>6 M</b> |
|            | $c$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $h$                          |     |     |     |                              |                              |     |                              |                              |  |     |     |     |            |            |          |                   |                   |            |
| $c$        | $P(X_{t+1} = c   X_t = c) =$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $P(X_{t+1} = c   X_t = h) =$ |     |     |     |                              |                              |     |                              |                              |  |     |     |     |            |            |          |                   |                   |            |
| $h$        | $P(X_{t+1} = h   X_t = c) =$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $P(X_{t+1} = h   X_t = h) =$ |     |     |     |                              |                              |     |                              |                              |  |     |     |     |            |            |          |                   |                   |            |
|            | $c$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $h$                          |     |     |     |                              |                              |     |                              |                              |  |     |     |     |            |            |          |                   |                   |            |
| $m$        | $P(m c) =$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | $P(m h) =$                   |     |     |     |                              |                              |     |                              |                              |  |     |     |     |            |            |          |                   |                   |            |
| $\neg m$   | $P(\neg m   c) =$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $P(\neg m   h) =$            |     |     |     |                              |                              |     |                              |                              |  |     |     |     |            |            |          |                   |                   |            |

**Q.3**

- a. Consider the following Bayesian network where A, B, C, and D are Boolean random variables. If we know that A is true, what is the probability of D being true. (Show all steps of calculation)



- b. Consider the following Bayesian network, where F = having the COVID and C = having flu. Fill the following joint probability table. (Show all steps of calculations)



| F     | C     | Joint Probability |
|-------|-------|-------------------|
| True  | True  |                   |
| True  | False |                   |
| False | True  |                   |
| False | False |                   |

**Q.4**

- a. Define reinforcement learning. Write the generalized equation for Q-learning for calculating Q-value,  $Q(s,a)$  and describe each of the parameter in equation.  
 b. Write any four correct differences between bagging and boosting.

**Q.5**

Consider the following two-dimensional sample,

|   | X  | Y  |
|---|----|----|
| A | 40 | 50 |
| B | 10 | 40 |
| C | 82 | 89 |
| D | 53 | 72 |
| E | 91 | 98 |

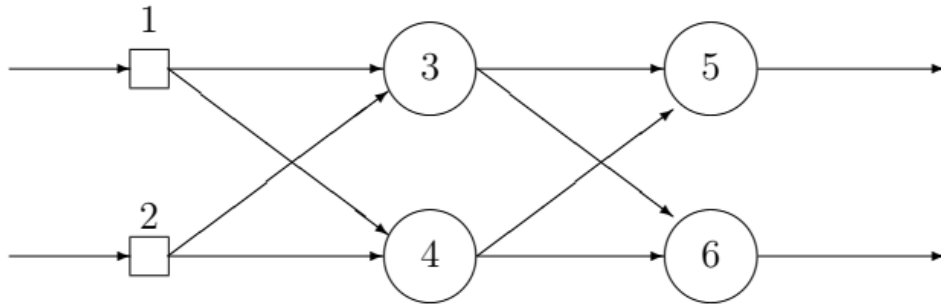
Use Principal Component Analysis (PCA) to reduce the dimension of above data from 2 to 1. Show all steps of calculation of covariance matrix, eigen values, normalized eigen vectors, and draw sample points in one dimension with respect to Principal Component 2. (Show all steps of calculation)

**3 + 3 M****2 + 2 M****6 M**

**Q. 6**

The following diagram represents a feed-forward neural network with one hidden layer:

**6 M**



A weight on connection between nodes  $i$  and  $j$  is denoted by  $w_{ij}$ , such as  $w_{13}$  is the weight on the connection between nodes 1 and 3. The following table lists all the weights in the network:

|               |               |
|---------------|---------------|
| $w_{13} = -2$ | $w_{35} = 1$  |
| $w_{23} = 3$  | $w_{45} = -1$ |
| $w_{14} = 4$  | $w_{36} = -1$ |
| $w_{24} = -1$ | $w_{46} = 1$  |

Each of the nodes 3, 4, 5 and 6 uses the following activation function:

$$\varphi(v) = \begin{cases} 1 & \text{if } v \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

where  $v$  denotes the weighted sum of a node. Each of the input nodes (1 and 2) can only receive binary values (either 0 or 1). Calculate the output of the network ( $y_5$  and  $y_6$ ) for each of the input patterns:

| Pattern | P1 | P2 | P3 | P4 |
|---------|----|----|----|----|
| Node 1  | 0  | 1  | 0  | 1  |
| Node 2  | 0  | 0  | 1  | 1  |

**Q. 7**

- a.** Write any two advantages and disadvantages of SVM?
- b.** Which of the following statements are true regarding Lasso regression? Give reason for it.
- Lasso's optimization problem can be stated as a quadratic program.
  - The cost function minimized by Lasso has points where its gradient is not well-defined, and the solution (minimum) is often at such a point.
  - Lasso often produces sparser results (more zero weights) than ridge regression.
  - A version of Lasso using a penalty term of  $\lambda \|w\|_{\ell_{0.5}}$  (that is, the  $\ell_{0.5}$ -norm) will be more inclined to produce sparse solutions than Lasso.
- c.** Which of the following algorithms can learn nonlinear decision boundaries? The decision trees use only axis-aligned splits. Give reason for it.
- A depth-five decision tree
  - Quadratic discriminant analysis (QDA)
  - AdaBoost with depth-one decision trees
  - Perceptron

**3 M**

\*\*\*\*\* ALL THE BEST \*\*\*\*\*

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|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|-----|-----|-----|------------------------------|------------------------------|-----|------------------------------|------------------------------|------------|
|            | <p>1<sup>st</sup> correct iteration =&gt; 3 M</p> <p>2<sup>nd</sup> correct iteration =&gt; 3 M</p> <p>Correct mean, variance and mixing coefficient of each cluster =&gt; 3 M</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                              |     |     |     |                              |                              |     |                              |                              |            |
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|            | $c$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | $h$                          |     |     |     |                              |                              |     |                              |                              |            |
| $c$        | $P(X_{t+1} = c   X_t = c) =$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $P(X_{t+1} = c   X_t = h) =$ |     |     |     |                              |                              |     |                              |                              |            |
| $h$        | $P(X_{t+1} = h   X_t = c) =$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $P(X_{t+1} = h   X_t = h) =$ |     |     |     |                              |                              |     |                              |                              |            |

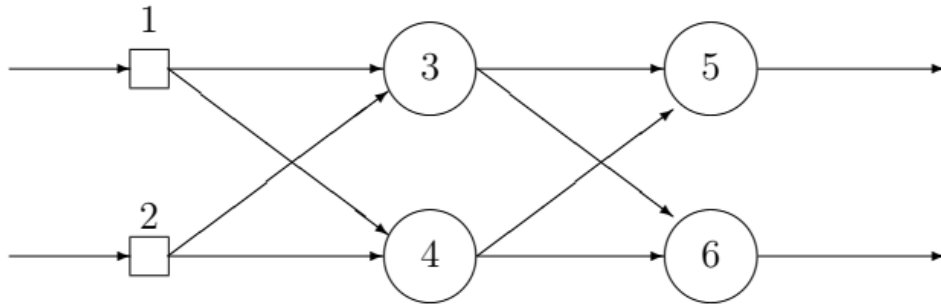
|            | <table> <tr> <td></td><td><math>c</math></td><td><math>h</math></td></tr> <tr> <td><math>m</math></td><td><math>P(m c) =</math></td><td><math>P(m h) =</math></td></tr> <tr> <td><math>\neg m</math></td><td><math>P(\neg m c) =</math></td><td><math>P(\neg m h) =</math></td></tr> </table> <p>c. What is the probability of producing the exact sequence <math>C, H, H, C, H</math> (from <math>t = 0</math> to <math>t = 4</math>)? Ignore evidence. Consider that there's a 50% chance Student's initial classification is Easy, <math>P(X_0 = c) = 0.5</math>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                | $c$ | $h$               | $m$  | $P(m c) =$                     | $P(m h) =$                     | $\neg m$ | $P(\neg m c) =$                | $P(\neg m h) =$                |       |      |     |       |                |                |                |                     |                     |  |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|-----|-------------------|------|--------------------------------|--------------------------------|----------|--------------------------------|--------------------------------|-------|------|-----|-------|----------------|----------------|----------------|---------------------|---------------------|--|
|            | $c$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | $h$                            |     |                   |      |                                |                                |          |                                |                                |       |      |     |       |                |                |                |                     |                     |  |
| $m$        | $P(m c) =$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $P(m h) =$                     |     |                   |      |                                |                                |          |                                |                                |       |      |     |       |                |                |                |                     |                     |  |
| $\neg m$   | $P(\neg m c) =$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | $P(\neg m h) =$                |     |                   |      |                                |                                |          |                                |                                |       |      |     |       |                |                |                |                     |                     |  |
|            | <p>a. 2M</p> <table> <tr> <td></td><td><math>c</math></td><td><math>h</math></td></tr> <tr> <td><math>c</math></td><td><math>P(X_{t+1} = c X_t = c) = 0.3</math></td><td><math>P(X_{t+1} = c X_t = h) = 0.7</math></td></tr> <tr> <td><math>h</math></td><td><math>P(X_{t+1} = h X_t = c) = 0.7</math></td><td><math>P(X_{t+1} = h X_t = h) = 0.3</math></td></tr> </table> <p>b. 2 M</p> <table> <tr> <td></td><td><math>c</math></td><td><math>h</math></td></tr> <tr> <td><math>m</math></td><td><math>P(m c) = 0.6</math></td><td><math>P(m h) = 0.8</math></td></tr> <tr> <td><math>\neg m</math></td><td><math>P(\neg m c) = 0.4</math></td><td><math>P(\neg m h) = 0.2</math></td></tr> </table> <p>c. 0.05145</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                | $c$ | $h$               | $c$  | $P(X_{t+1} = c X_t = c) = 0.3$ | $P(X_{t+1} = c X_t = h) = 0.7$ | $h$      | $P(X_{t+1} = h X_t = c) = 0.7$ | $P(X_{t+1} = h X_t = h) = 0.3$ |       | $c$  | $h$ | $m$   | $P(m c) = 0.6$ | $P(m h) = 0.8$ | $\neg m$       | $P(\neg m c) = 0.4$ | $P(\neg m h) = 0.2$ |  |
|            | $c$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | $h$                            |     |                   |      |                                |                                |          |                                |                                |       |      |     |       |                |                |                |                     |                     |  |
| $c$        | $P(X_{t+1} = c X_t = c) = 0.3$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | $P(X_{t+1} = c X_t = h) = 0.7$ |     |                   |      |                                |                                |          |                                |                                |       |      |     |       |                |                |                |                     |                     |  |
| $h$        | $P(X_{t+1} = h X_t = c) = 0.7$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | $P(X_{t+1} = h X_t = h) = 0.3$ |     |                   |      |                                |                                |          |                                |                                |       |      |     |       |                |                |                |                     |                     |  |
|            | $c$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | $h$                            |     |                   |      |                                |                                |          |                                |                                |       |      |     |       |                |                |                |                     |                     |  |
| $m$        | $P(m c) = 0.6$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | $P(m h) = 0.8$                 |     |                   |      |                                |                                |          |                                |                                |       |      |     |       |                |                |                |                     |                     |  |
| $\neg m$   | $P(\neg m c) = 0.4$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | $P(\neg m h) = 0.2$            |     |                   |      |                                |                                |          |                                |                                |       |      |     |       |                |                |                |                     |                     |  |
| <b>Q.3</b> | <p>a. Consider the following Bayesian network where A, B, C, and D are Boolean random variables. If we know that A is true, what is the probability of D being true. (Show all steps of calculation)</p> <div style="text-align: center;"> <pre> graph TD     A((A)) --&gt; B((B))     A((A)) --&gt; C((C))     B((B)) --&gt; D((D))     C((C)) --&gt; D((D)) </pre> <p> <math>P(A) = 0.75</math><br/> <math>P(B A) = 0.2</math><br/> <math>P(B \neg A) = 0.5</math><br/> <math>P(C A) = 0.7</math><br/> <math>P(C \neg A) = 0.25</math><br/> <math>P(D B \wedge C) = 0.3</math><br/> <math>P(D B \wedge \neg C) = 0.25</math><br/> <math>P(D \neg B \wedge C) = 0.1</math><br/> <math>P(D \neg B \wedge \neg C) = 0.35</math> </p> </div> <p>b. Consider the following Bayesian network, where F = having the COVID and C = having flu. Fill the following join probability table. (Show all steps of calculations)</p> <div style="text-align: center;"> <pre> graph LR     F((F)) --&gt; C((C)) </pre> <p> <math>P(F) = 0.1</math><br/> <math>P(C F) = 0.8</math><br/> <math>P(C \neg F) = 0.3</math> </p> <table border="1"> <thead> <tr> <th>F</th><th>C</th><th>Joint Probability</th></tr> </thead> <tbody> <tr> <td>True</td><td>True</td><td></td></tr> <tr> <td>True</td><td>False</td><td></td></tr> <tr> <td>False</td><td>True</td><td></td></tr> <tr> <td>False</td><td>False</td><td></td></tr> </tbody> </table> </div> | F                              | C   | Joint Probability | True | True                           |                                | True     | False                          |                                | False | True |     | False | False          |                | <b>3 + 3 M</b> |                     |                     |  |
| F          | C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Joint Probability              |     |                   |      |                                |                                |          |                                |                                |       |      |     |       |                |                |                |                     |                     |  |
| True       | True                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                |     |                   |      |                                |                                |          |                                |                                |       |      |     |       |                |                |                |                     |                     |  |
| True       | False                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                |     |                   |      |                                |                                |          |                                |                                |       |      |     |       |                |                |                |                     |                     |  |
| False      | True                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                |     |                   |      |                                |                                |          |                                |                                |       |      |     |       |                |                |                |                     |                     |  |
| False      | False                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                |     |                   |      |                                |                                |          |                                |                                |       |      |     |       |                |                |                |                     |                     |  |

|             | <p>a. 0.197<br/>b.</p> <table border="1"> <thead> <tr> <th>F</th><th>C</th><th>Joint Probability</th></tr> </thead> <tbody> <tr> <td>True</td><td>True</td><td>0.08</td></tr> <tr> <td>True</td><td>False</td><td>0.02</td></tr> <tr> <td>False</td><td>True</td><td>0.27</td></tr> <tr> <td>False</td><td>False</td><td>0.63</td></tr> </tbody> </table>                                                                                                                                                                                                                                                                                                                                                                         | F                 | C | Joint Probability | True     | True | 0.08 | True     | False | 0.02 | False    | True | 0.27 | False    | False | 0.63 |          |    |    |            |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|---|-------------------|----------|------|------|----------|-------|------|----------|------|------|----------|-------|------|----------|----|----|------------|
| F           | C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Joint Probability |   |                   |          |      |      |          |       |      |          |      |      |          |       |      |          |    |    |            |
| True        | True                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 0.08              |   |                   |          |      |      |          |       |      |          |      |      |          |       |      |          |    |    |            |
| True        | False                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 0.02              |   |                   |          |      |      |          |       |      |          |      |      |          |       |      |          |    |    |            |
| False       | True                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 0.27              |   |                   |          |      |      |          |       |      |          |      |      |          |       |      |          |    |    |            |
| False       | False                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 0.63              |   |                   |          |      |      |          |       |      |          |      |      |          |       |      |          |    |    |            |
| <b>Q.4</b>  | <p>a. Define reinforcement learning. Write the generalized equation for Q-learning for calculating Q-value, <math>Q(s,a)</math> and describe each of the parameter in equation.<br/>b. Write any four correct differences between bagging and boosting.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>2 + 2 M</b>    |   |                   |          |      |      |          |       |      |          |      |      |          |       |      |          |    |    |            |
|             | <p>a. Correct definition with Q-learning formula and description of each parameter =&gt; 2 M<br/>b. 0.5 marks for each correct difference.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                   |   |                   |          |      |      |          |       |      |          |      |      |          |       |      |          |    |    |            |
| <b>Q. 5</b> | <p>Consider the following two-dimensional sample,</p> <table border="1"> <thead> <tr> <th></th><th>X</th><th>Y</th></tr> </thead> <tbody> <tr> <td><b>A</b></td><td>40</td><td>50</td></tr> <tr> <td><b>B</b></td><td>10</td><td>40</td></tr> <tr> <td><b>C</b></td><td>82</td><td>89</td></tr> <tr> <td><b>D</b></td><td>53</td><td>72</td></tr> <tr> <td><b>E</b></td><td>91</td><td>98</td></tr> </tbody> </table> <p>Use Principal Component Analysis (PCA) to reduce the dimension of above data from 2 to 1. Show all steps of calculation of covariance matrix, eigen values, normalized eigen vectors, and draw sample points in one dimension with respect to Principal Component 2. (Show all steps of calculation)</p> |                   | X | Y                 | <b>A</b> | 40   | 50   | <b>B</b> | 10    | 40   | <b>C</b> | 82   | 89   | <b>D</b> | 53    | 72   | <b>E</b> | 91 | 98 | <b>6 M</b> |
|             | X                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Y                 |   |                   |          |      |      |          |       |      |          |      |      |          |       |      |          |    |    |            |
| <b>A</b>    | 40                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 50                |   |                   |          |      |      |          |       |      |          |      |      |          |       |      |          |    |    |            |
| <b>B</b>    | 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 40                |   |                   |          |      |      |          |       |      |          |      |      |          |       |      |          |    |    |            |
| <b>C</b>    | 82                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 89                |   |                   |          |      |      |          |       |      |          |      |      |          |       |      |          |    |    |            |
| <b>D</b>    | 53                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 72                |   |                   |          |      |      |          |       |      |          |      |      |          |       |      |          |    |    |            |
| <b>E</b>    | 91                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 98                |   |                   |          |      |      |          |       |      |          |      |      |          |       |      |          |    |    |            |
|             | <p>Mean =&gt; 1 M<br/>Covariance matrix =&gt; 1 M<br/>Eigen Value =&gt; 1 M<br/>Eigen Vector =&gt; 1 M<br/>Normalized Eigen Vector =&gt; 1 M<br/>Plot =&gt; 1 M</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                   |   |                   |          |      |      |          |       |      |          |      |      |          |       |      |          |    |    |            |

**Q. 6**

The following diagram represents a feed-forward neural network with one hidden layer:

**6 M**



A weight on connection between nodes  $i$  and  $j$  is denoted by  $w_{ij}$ , such as  $w_{13}$  is the weight on the connection between nodes 1 and 3. The following table lists all the weights in the network:

|               |               |
|---------------|---------------|
| $w_{13} = -2$ | $w_{35} = 1$  |
| $w_{23} = 3$  | $w_{45} = -1$ |
| $w_{14} = 4$  | $w_{36} = -1$ |
| $w_{24} = -1$ | $w_{46} = 1$  |

Each of the nodes 3, 4, 5 and 6 uses the following activation function:

$$\varphi(v) = \begin{cases} 1 & \text{if } v \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

where  $v$  denotes the weighted sum of a node. Each of the input nodes (1 and 2) can only receive binary values (either 0 or 1). Calculate the output of the network ( $y_5$  and  $y_6$ ) for each of the input patterns:

| Pattern | P1 | P2 | P3 | P4 |
|---------|----|----|----|----|
| Node 1  | 0  | 1  | 0  | 1  |
| Node 2  | 0  | 0  | 1  | 1  |

(1,1) (0,1) (1,0) (1,1) => 1.5 M for each correct answer with steps

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |     |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| Q. 7 | <p><b>a.</b> Write any two advantages and disadvantages of SVM?</p> <p><b>b.</b> Which of the following statements are true regarding Lasso regression? Give reason for it.</p> <ul style="list-style-type: none"> <li>i. Lasso's optimization problem can be stated as a quadratic program.</li> <li>ii. The cost function minimized by Lasso has points where its gradient is not well-defined, and the solution (minimum) is often at such a point.</li> <li>iii. Lasso often produces sparser results (more zero weights) than ridge regression.</li> <li>iv. A version of Lasso using a penalty term of <math>\lambda \ w\ _{\ell_{0.5}}</math> (that is, the <math>\ell_{0.5}</math>-norm) will be more inclined to produce sparse solutions than Lasso.</li> </ul> <p><b>c.</b> Which of the following algorithms can learn nonlinear decision boundaries? The decision trees use only axis-aligned splits. Give reason for it.</p> <ul style="list-style-type: none"> <li>i. A depth-five decision tree</li> <li>ii. Quadratic discriminant analysis (QDA)</li> <li>iii. AdaBoost with depth-one decision trees</li> <li>iv. Perceptron</li> </ul> | 3 M |
|      | <p>a. 0.5 M for each advantage and disadvantage</p> <p>b. iii, iv,<br/>ii is true. This is due to the square-shaped isocontours introduced by the '1 norm being more likely to intersect other isocontours on axes than the circular isocontours introduced by the '2 norm.<br/>iv is true — intuitively, the smaller the p we select for our p-norm, the pointier the penalty's isocontours become, making us more likely to choose a sparse weight vector.</p> <p>c. ii, iv</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |     |

\*\*\*\*\* ALL THE BEST \*\*\*\*\*